SEMI-FLEX DESIGN GUIDELINES 1.1

NCAB GROUP
Integrated PCB Production

Build capabilities 1. STRUCTURE TYPES

ASYMMETRICAL

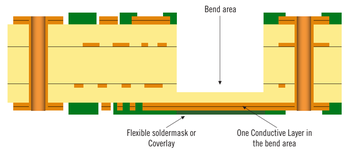
Asymmetrical Semi-Flex PCB is made by removing material from a certain portion on one side of the rigid printed board to form a bendable area. This structure is preferred in design and process for Semi-Flex.

SYMMETRICAL

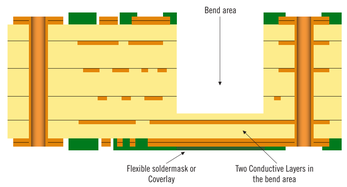
Higher cost and more complex – not recommended

Symmetrical Semi-Flex PCB is made by removing material from a certain portion on two sides of the rigid printed board to form a bendable area. Symmetrical structure is also called 'Regal Flex' where the bending layer is situated within the center of the rigid construction. Yet this structure adds cost due to the double Z-axis routing and twice uncovering necessary.

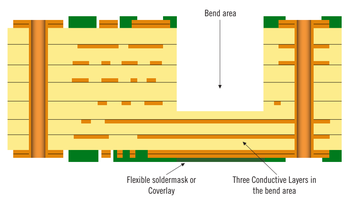
Build capabilities 2. CONDUCTIVE LAYERS IN THE BEND AREA TYPES



PREFERRED



POSSIBLE



NOT RECOMMENDED

Material capabilities

- The common FR4 or FR4.1 material, including mid Tg and High Tg material would be used for semi-flex accordingly.
- Normal Tg (<=140°C) material is with lower flexural strength and it is more flexible for bending, but it will be limited for lead-free process. Possible 2-4L only.
- There are some specific type of flexible material but with limitation, check with our FAE for details.
- Thinner glass fiber like 106 and 1080 or similar thin glass fiber will be suggested inside the stack up and at bending layer. (Symmetrical construction should be used.)
- 2116 and 7628 PP is not recommended used at bending area.
- Flexible soldermask or coverlay would be used at bending area.

Getting it right from the start

Nothing affects the PCB's cost and quality as much as the initial design. As modern electronic products are expected to offer more and more advanced functions, while the products themselves are becoming smaller and smaller, this puts greater demands on the PCB design and the importance of making the right choices at the design stage.

More than 30% of the Gerber data packs we receive do have some issues, generally, ambiguous information, errors, design rule conflicts, missing information and contradictions between the data and specifications. To prevent that it gets wrong from the start, we have put together these design guidelines, to use as a checklist…

When using combinations of parameters, you should always consult your local NCAB technical contact person.

Design tips

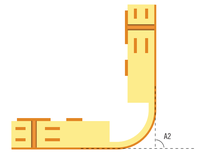
1 **Semi-Flex can be** used for 'Flex-To-Install' static applications, or applications that have a VERY limited or controlled number of bends. 'Dynamic Flex' usage or multiple flexing increases risk of cracks in FR4/copper.

2 **For dynamic use** materials(FPC+ coverlay) with similar properties.

3 **Estimation of minimum bend length** (marked 'G' in below figures as shown under Process and technology capabilities). Many factors can play into a semi-flex circuit's ability to withstand the bends required by the application. Some of these factors include, but are not limited to

a) Base material type and its flexibility
b) Bend area thickness
c) Bend radius
d) Copper type(RA, ED, etc.), thickness and elongation
e) Static applications requires specific bending angle& bending cycles
f) Conductive layers in bend area and stress concentration features.

Since no formula or calculation can possibly predict every possible construction and usage, it is recommended that the circuit design follow below formula to estimate the basic bend length, which is given by:
G= (πR (A2) ÷180) + 4mm
where:
G=minimum bend length
R=basic inner bend radius
A2=the bending angle, in degrees.
4mm is from experimental empirical value for safety compensation.



Detailed calculations about bend radius should refer to the basic empirical formulas which can be seen as below:

CONDUCTIVE LAYERS IN THE BEND AREA	MINIMUM BEND RADIUS (R) FOR ≤ 90 DEGREE BENDING	
	STANDARD	ADVANCED
1 layer	≥ 5mm	≥ 3mm
2 layer	≥ 7mm	≥ 5mm

It is preferable that a semi-flex PCB be allowed to follow its own natural bend. Bends greater than 90°and containing a small bend radius should be avoided.

4 **For 180 degree bending:**
For improved strength and reliability, a 2 x 90 degree bending (Fig 1) is suggested to be used instead of 180 degree bending.

For 180degree bending, radius >=5mm is recommended.

It is recommended to use a thin FR4 core 0.075-0.1mm (Fig. 2), this will give more flexibility for bending. Due to the use of de-capping process, the remaining surface will be smoother and better thickness tolerance. Only 1L copper is recommended.

During the bending operation, it is suggested to use a support tool (bending die) (Fig. 3). Make sure this mandrel fully supports the semi-flex area during the bend.

For reliability and avoid fractures in the material under bending, we recommend copper fill in the bending area. Add dummy copper where needed, in cooperation with NCAB FAE. See point 8.

Bending in Warp direction of the glass fiber will improve bending performance. So to design all bending zones in same direction is recommended.

Fig 1

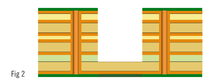


Fig 2

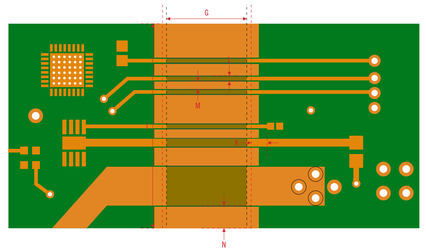
Fig 3

5 **Maximum bending cycles**
Although SemiFlex is a bend to install technology, repeated bend cycles may be possible. Maximum number of bend cycles must be discussed with your local NCAB FAE. The required bend cycles shall be clearly written in the procurement documentation.

6 **Selection of Flexible Soldermask or Coverlay**
Generally, flexible soldermask is recommended for coating outer layer copper feature only in bend area when less than 10 bending cycles and where bending strength is not high. If bending application requires more than 10 bending cycles and high flexural strength, it is recommended that using Coverlay.

7 **The bending area** should be facing outwards. Facing inwards would greatly increase the mechanical stress that may weaken the structural integrity of the PCB. The copper layer shall be on the outer layer / outside in the bend area. See Fig 1-3 in Design Tips 4.

8 **If the bend area contains only a single conductive / copper layer**, then It is recommended to add copper fill in the bending area to even out the copper distribution in the bending area. This will distribute the forces and minimize the risk of cracks. An example can be seen in the graphic below where the yellow colour represents added copper. If the design contains more than one conductive / copper layer in the bend area, then it is recommended to contact your local NCAB FAE to discuss the design.

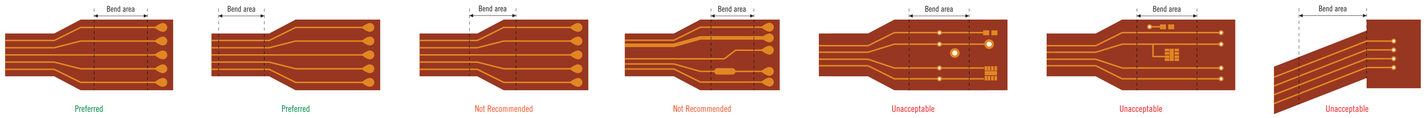


If you have questions about these guidelines or other design details please contact your local NCAB office for more details.

Circuitry design

Preferred | Acceptable | Unacceptable

Eliminate stress concentrated copper corners, see the three figures above.



Preferred | Preferred | Not Recommended | Not Recommended | Unacceptable | Unacceptable | Unacceptable

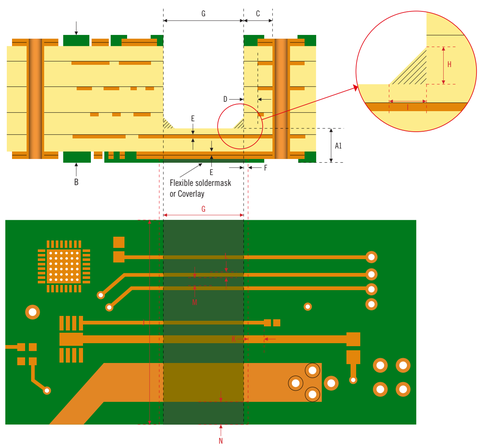
Copper tracks shall be perpendicular to the bend, so as to prevent from tearing and scratching. Do not design for bend areas where tracks are curved or angled, like the three figures above.

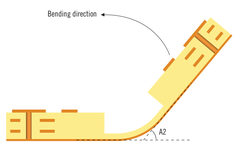
If possible tracks shall also evenly spread across the bend area and shall maintain a constant width.

Vias and PTHs in bend areas should be avoided.

Without additional plated metals exposure outside.

Process and technology capabilities





SYMBOL	DESCRIPTION		STANDARD	ADVANCED
A	A1 Remaining thickness in bend area thickness (according to conductive layers in bend area)	1 conductive layer in bend area	0.20-0.30mm	0.15-0.35mm
		2 conductive layers in bend area	0.25-0.35mm	0.20-0.40mm
	For advanced designs where the minimum thickness in the bend area is 0.15mm, please discuss the design with NCAB technical staff in order to best define nominal thickness and tolerances		±0.075mm	±0.05mm
	A2 Bending angle & Bending cycles		0~90°& < 5cycles	0~180°& < 20cycles
B	Board thickness in rigid part & rigid portion thickness		1.0~2.4mm	0.8~2.4mm
C	Spacing between PTH edge to rigid/bendable transition		≥ 0.8mm	≥ 0.5mm
D	Spacing between I/L copper feature edge to rigid/bendable transition		≥ 0.4mm	≥0.25mm
E	Base copper thickness on bend area		1/3 oz~Hoz	1/3 oz~1oz
F	Flexible soldermask access conventional soldermask overlap(min 2 locations)		≥ 0.3mm	≥ 0.25mm
	Coverlay access conventional soldermask overlap(min 2 locations)		≥ 0.25mm	≥ 0.1mm
G	Bend length(according to conductive layers in bend area and bending angle)	1 conductive layer in bend area	≥15.0mm (bending angle 30°)	≥6.5mm(bending angle 30°)
		1 conductive layers in bend area	≥18.0mm(bending angle 90°)	≥9.0mm(bending angle 90°)
		2 conductive layers in bend area	≥18.0mm(bending angle 90°)	≥12mm(bending angle 90°)
H	Transition corner height (depending on glue and mechanical routing)	Eccobond / glue at the transition corner	≤B-A1	≤B-A1
		Mechanical depth control milling (Common)	0.3~1.0mm	0.2~1.0mm
I	Transition corner length(depending on glue and mechanical routing)	Eccobond / glue at the transition corner	0.5~2.0mm	0.5~1.5mm
		Mechanical depth control milling (Common)	0.3~1.0mm	0.2~1.0mm
J	Bend area width		≥ 5.0mm	≥3.0 mm
K	Spacing between O/L copper edge to rigid/bendable transition		≥ 1.0mm	≥ 0.5mm
L/M	Min. trace width(L) & spacing(M) in bend area (according to base copper thickness)	0.5oz	0.102/0.102mm	0.889/0.889mm
		1oz	0.127/0.127mm	0.102/0.127mm
N	Space between bend area copper feature edge to outline		≥ 0.40mm	≥ 0.20mm

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